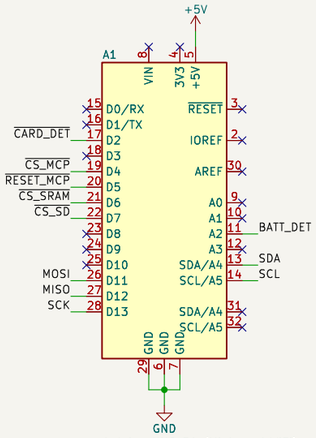
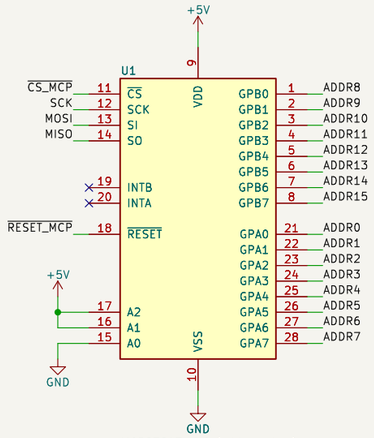
FUJISOKU SRAM Utility — schematic

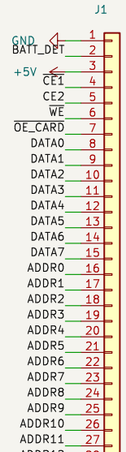
Single 5 V supply from the Arduino 5V pin. All parts are 5 V, no level shifting.
SPI (D13 SCK, D12 MISO, D11 MOSI) is shared by both MCP23S17, the SPI SRAM and the SD card module.
Labels with the same name are connected. Source: docs/hardware.md, FUJISOKU_SRAM_COPY/config.h (v3.0).



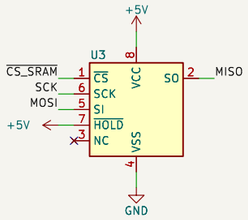
Arduino UNO R4 (Minima / WiFi)
D2 = /CARD DET (F37), INPUT_PULLUP, L = card in
A2 = BATT-DET (F2), analog in
D3, D8–D10, A0, A1, A3: not used



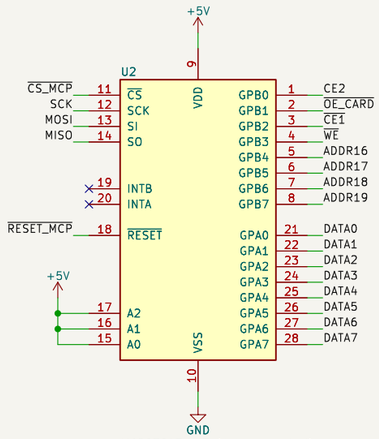
MCP23S17 ch6 (address bus)
Hardware address A2:A0 = 110 (ch6)
GPA = A0–A7, GPB = A8–A15



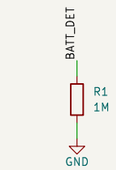
FUJISOKU card slot (F1–F38)
Pin n = card pin Fn. F32–F35 = A16–A19 (U2 GPB4–GPB7).
F3/F36 (Vcc) and F1/F38 (GND) are joined inside the card.
Slot made with 2.54 mm <–> 1.27 mm adapters (docs/hardware.md 12.1).



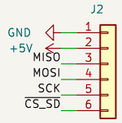
23LC512 SPI SRAM (optional)
USE_SPI_SRAM (auto-detected)



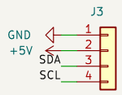
MCP23S17 ch7 (data + control)
Hardware address A2:A0 = 111 (ch7)
GPA = D0–D7, GPB0 CE2, GPB1 /OE, GPB2 /CE1, GPB3 /WE,
GPB4 A16, GPB5 A17, GPB6 A18, GPB7 A19



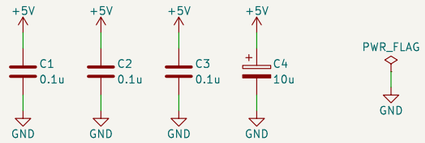
BATT-DET pull-down: no battery 0.24 V, new battery 2.75 V



SD card module, 5 V type (optional)
USE_SD. 3.3 V regulator + level shifter on module
(pin order differs by module)



LCD 20x4, PCF8574 I2C backpack 0x27 (optional)
USE_LCD (auto-detected)



Bypass capacitors: C1 U1, C2 U2, C3 U3 (0.1u ceramic). C4 10u bulk near the card (card supply)

Generated by hardware/kicad/gen_schematic.py from docs/hardware.md
Arduino UNO R4 + MCP23S17 x 2 reader / writer for FUJISOKU SRAM memory cards
github.com/cheeky-smokey/FUJISOKU-SRAM-Utility

Sheet: /		
File: FUJISOKU_SRAM_Utility.kicad_sch		
Title: FUJISOKU SRAM Utility		
Size: A3	Date: 2026-11-14	**Rev: 3.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1